

Internship Report
On
Python Full Stack Development

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: CS-23411016
Name of Student: MOHD KAIF
Batch: 2026-27

CANDIDATE'S DECLARATION

I, MOHD KAIF, do hereby declare that the internship report titled “Python Full Stack Development” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: MOHD KAIF

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank EduSkills Academy and AICTE for providing the opportunity to undergo this Virtual Internship Program under the National Internship Portal, and IILM University, Greater Noida, for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

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TABLE OF CONTENTS

Candidate's Declaration	i
Acknowledgement.....	ii
Table of Contents.....	iii
List of Figures	iv
List of Tables	iv
Internship Completion Certificate	v
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	1
4.3 Problem Statement	1
4.4 Project Objectives	1
4.5 Scope of the Project	2
4.6 Technologies and Tools Used.....	2
4.7 System Architecture	3
4.8 Database Design (ER Diagram).....	3
4.9 Methodology.....	4
4.10 Sample Code.....	5
4.11 Screenshots	6
4.12 Expected Outcomes / Learning Outcomes	6
4.13 Conclusion and Future Scope	6
4.14 Certificates of Completion and Communication Proof	6
5. Bibliography / References.....	8

LIST OF FIGURES

Fig. 4.1 System Architecture Diagram.....	3
Fig. 4.2 Entity-Relationship (ER) Diagram.....	4
Fig. 4.3 Week-wise Methodology Flow	4

LIST OF TABLES

Table 4.1 Technologies and Tools Used	3
Table 4.2 Week-wise Internship Schedule	4
Table 4.3 Weekly Progress / Internship Log	8

INTERNSHIP COMPLETION CERTIFICATE



PRESENTED TO

Mohd Kaif

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Python Full Stack Development With Project

During the internship, the learner demonstrated proficiency in:

- Introduction to Full Stack Development & Project Setup & Python Fundamentals for Backend Development
- Introduction to FastAPI for Backend APIs & Data Validation and Modeling with Pydantic
- Introduction to React and Frontend Fundamentals & React Hooks and State Management
- Building Reusable React Components and Styling & Integrating FastAPI Backend with React Frontend
- Database Integration with FastAPI (SQLAlchemy) & Advanced FastAPI Features and Best Practices
- Advanced React Patterns and State Management (Redux/Zustand) & Frontend Routing and Navigation
- Building Forms and Handling User Input in React & Testing Your Full Stack Application & Deployment Strategies for Full Stack Applications
- API Design Principles and Best Practices & Frontend Best Practices and Performance Optimization & Capstone Project: E-commerce Platform with User Authentication and Product Catalog
- **Project: In-Memory Cache with TTL Lite**

This certification recognizes the successful completion of the internship requirements and project work.



Issued on: Jun 12, 2026
Certificate ID: 2026-CD94DBF412

Mihir Swain
General Manager
(Learning & Development)
EduSkills

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during an 8-week AICTE – EduSkills Virtual Internship in Python Full Stack Development, undertaken from April 2026 to June 2026 as part of the B.Tech (Computer Science and Engineering) curriculum at IILM University, Greater Noida. The internship followed a structured, week-wise curriculum covering backend development with Python and FastAPI, frontend development with React, database integration using SQLAlchemy, and modern software engineering practices such as testing and deployment. The program concluded with the design and development of a capstone full-stack e-commerce application and a supplementary mini-project, an in-memory cache with time-to-live (TTL) support.

4.2 Organization Profile

The internship was conducted under the AICTE – EduSkills Virtual Internship Program, a national initiative supported by the All India Council for Technical Education (AICTE) under its National Internship Portal, in collaboration with EduSkills Academy. EduSkills Academy is an ISO-certified skilling organisation that partners with academia and industry to deliver structured, industry-relevant virtual internship programs aimed at enhancing the employability of engineering students across India.

4.3 Problem Statement

Traditional academic curricula often provide limited hands-on exposure to the complete lifecycle of building and deploying a full-stack web application. Students frequently learn backend and frontend concepts in isolation, without experiencing how they integrate into a single, production-style system. This internship was designed to address that gap by requiring the design, development, and integration of a complete full-stack application — combining a Python-based backend, a React-based frontend, a relational database, authentication, and deployment considerations — within a structured eight-week timeline.

4.4 Project Objectives

The key objectives of the internship and the associated capstone project were:

- To gain proficiency in Python fundamentals, including data types, functions, and object-oriented programming, as a foundation for backend development.
- To design and build RESTful backend APIs using the FastAPI framework, with request validation and data modelling using Pydantic.
- To develop a responsive, component-based frontend using React, including hooks, state management, and routing.
- To integrate the frontend and backend, enabling dynamic data exchange through API calls.
- To design and integrate a relational database using SQLAlchemy ORM for persistent data storage and CRUD operations.
- To apply advanced React patterns and state management solutions such as Redux/Zustand for scalable frontend architecture.
- To implement testing strategies and understand deployment techniques for full-stack applications.
- To build a capstone e-commerce platform with user authentication and a product catalog, and a supporting in-memory cache utility with TTL support.

4.5 Scope of the Project

The scope of the internship project covers the end-to-end development of a full-stack e-commerce web application. On the backend, the scope includes API design, request handling, data validation, and database integration. On the frontend, the scope includes building reusable UI components, managing application state, handling user input through forms, and implementing client-side routing and navigation. The scope also extends to a supplementary systems-level mini-project — an in-memory cache with TTL (time-to-live) support — which reinforces backend fundamentals such as data structures and expiry management. Deployment considerations and testing practices were studied but implemented at a proof-of-concept level, appropriate to the scale of an academic internship.

4.6 Technologies and Tools Used

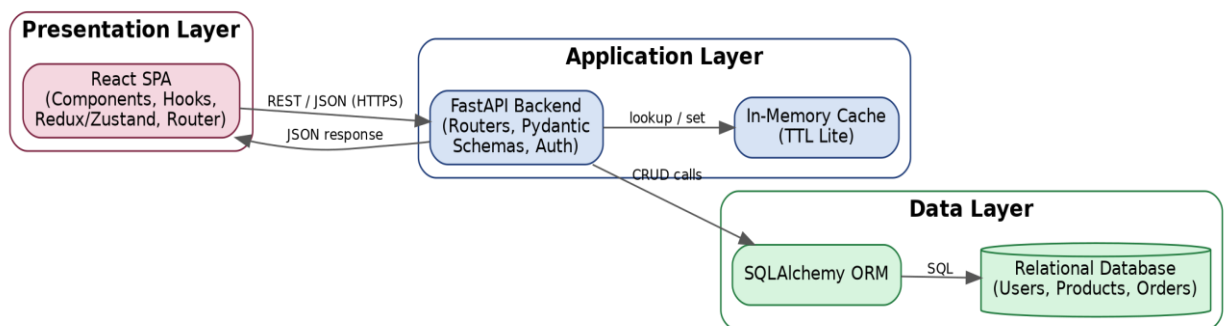
Table 4.1 summarises the primary technologies and tools used across the backend, frontend, database, and deployment layers of the project.

Table 4.1: Technologies and Tools Used

Category	Technology / Tool
Backend Language & Framework	Python, FastAPI
Data Validation	Pydantic
Frontend Library	React (Hooks, Components, Routing)
State Management	Redux / Zustand
Database & ORM	SQLAlchemy (relational database)
Testing	Backend and frontend testing frameworks
Deployment	Docker, cloud deployment platforms
Version Control	Git / GitHub

4.7 System Architecture

The capstone application follows a three-tier full-stack architecture, illustrated in Fig. 4.1. The presentation layer (React) communicates with the application layer (FastAPI) over REST/JSON, which in turn accesses the data layer (SQLAlchemy ORM over a relational database). An in-memory TTL cache sits alongside the application layer to reduce repeated database lookups for frequently accessed data.

**Fig. 4.1: System Architecture Diagram**

- **Presentation Layer:** A React single-page application responsible for rendering the UI, managing client-side state, and handling routing and navigation.
- **Application Layer:** A FastAPI backend exposing RESTful endpoints for authentication, product catalog management, and order processing, with validation via Pydantic models, and a TTL cache for hot-path lookups.
- **Data Layer:** A relational database accessed through the SQLAlchemy ORM, storing user, product, and order data.

4.8 Database Design (ER Diagram)

Fig. 4.2 shows the relational schema designed for the capstone e-commerce platform. A user can place multiple orders; each order consists of one or more order items, and each order item references a product from the catalog.

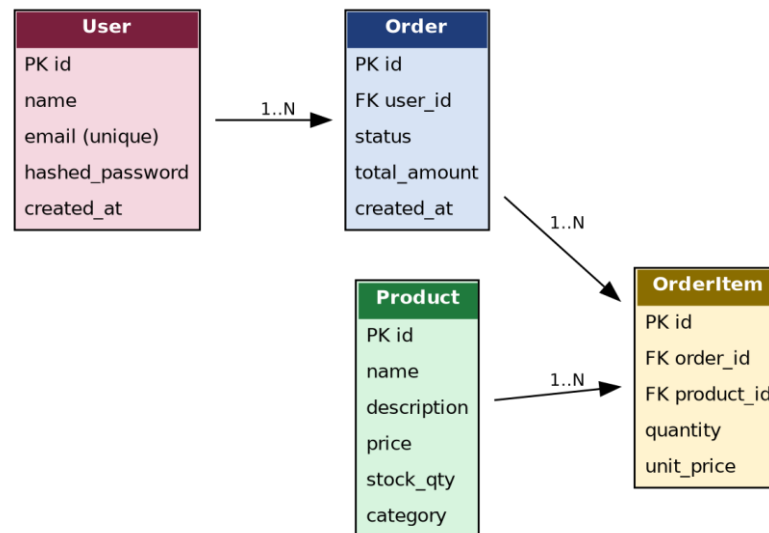


Fig. 4.2: Entity-Relationship (ER) Diagram

4.9 Methodology

The project was developed incrementally over the eight weeks of the internship, following the week-wise curriculum structure summarised in Table 4.2 and Fig. 4.3. Backend fundamentals were covered first, followed by API development and data validation, then frontend fundamentals, database integration, advanced state management, testing/deployment, and finally the capstone build.

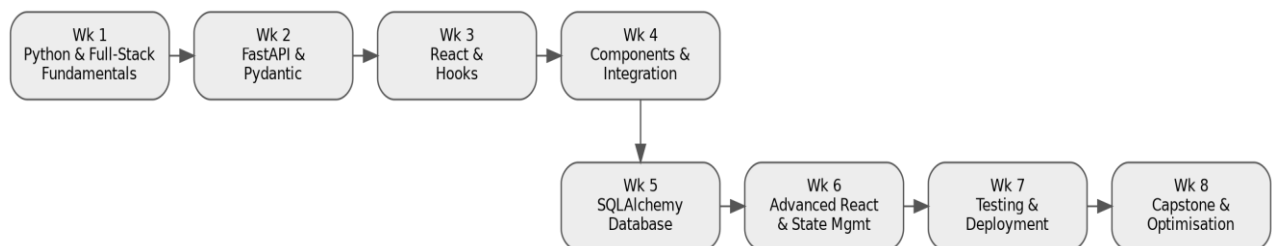


Fig. 4.3: Week-wise Methodology Flow

Table 4.2: Week-wise Internship Schedule

Week	Module	Description
1	Introduction to Full Stack & Python Fundamentals	Full-stack concepts, environment setup, Python data types, functions, OOP, backend basics.
2	FastAPI & Data Validation (Pydantic)	Backend APIs with FastAPI – request handling, routing, dependency injection, validation.

Week	Module	Description
3	React Fundamentals & Hooks	React basics – components, JSX, props, useState, useEffect, dynamic UIs.
4	Reusable Components & Frontend-Backend Integration	Reusable React components integrated with the FastAPI backend via API calls.
5	Database Integration with SQLAlchemy	SQLAlchemy ORM, CRUD operations, models, relational database connectivity.
6	Advanced React & State Management	Advanced patterns, React Router, Redux/Zustand for scalable applications.
7	Testing & Deployment	Backend/frontend testing strategies; deployment with Docker and cloud platforms.
8	API Design, Frontend Optimisation & Capstone Project	Full e-commerce application – FastAPI backend, database, auth, React frontend.

4.10 Sample Code

The following snippets illustrate two representative pieces of the implementation: a FastAPI route handler used in the capstone e-commerce backend, and the core logic of the In-Memory Cache with TTL Lite mini-project.

(a) FastAPI Route – Creating a Product

```
@router.post("/products", response_model=ProductOut, status_code=201)
def create_product(payload: ProductCreate, db: Session = Depends(get_db)):
    product = Product(**payload.model_dump())
    db.add(product)
    db.commit()
    db.refresh(product)
    return product
```

(b) In-Memory Cache with TTL Lite – Core Class

```
class TTLCache:
    """A minimal in-memory cache with per-key time-to-live (TTL)."""

    def __init__(self, default_ttl: float = 60.0):
        self._store: dict[str, tuple[Any, float]] = {}
        self.default_ttl = default_ttl

    def set(self, key: str, value: Any, ttl: float | None = None) -> None:
        expiry = time.monotonic() + (ttl or self.default_ttl)
        self._store[key] = (value, expiry)

    def get(self, key: str) -> Any | None:
        item = self._store.get(key)
        if item is None:
            return None
        value, expiry = item
        if time.monotonic() > expiry:
            del self._store[key]      # lazy expiry
            return None
        return value

    def delete(self, key: str) -> None:
        self._store.pop(key, None)
```

The cache uses lazy expiry: rather than running a background sweep, an expired key is detected and removed the next time it is accessed, keeping the implementation simple and dependency-free while still bounding memory growth for short-lived entries.

4.11 Screenshots

This section is reserved for screenshots of the running application (e.g. login page, product catalog, cart/checkout flow, and API documentation generated by FastAPI at /docs). Insert screenshots captured from your local run of the project here before final submission.

4.12 Expected Outcomes / Learning Outcomes

Upon completion of the internship, the following outcomes were achieved:

- A working understanding of building and consuming RESTful APIs using FastAPI.
- Practical experience in building responsive, component-based user interfaces using React.
- The ability to integrate a frontend application with a backend API and a relational database.
- A functional capstone e-commerce platform demonstrating user authentication and product catalog management.
- A working in-memory cache utility implementing TTL-based expiry (“In-Memory Cache with TTL Lite”).
- Familiarity with testing and deployment strategies for full-stack applications.

4.13 Conclusion and Future Scope

The internship provided practical, end-to-end exposure to full-stack web development — from backend API design and database modelling to frontend UI development and state management — within a structured eight-week program. The resulting capstone e-commerce platform and the In-Memory Cache with TTL Lite project demonstrate the core competencies expected of a full-stack developer. As future scope, the platform could be extended with features such as payment gateway integration, role-based access

control, containerised deployment on a cloud platform, and replacing the in-memory cache with a distributed cache (e.g. Redis) for multi-instance deployments.

4.14 Certificates of Completion and Communication Proof

The following pages contain the Internship Offer Letter issued by EduSkills Academy under the AICTE National Internship Portal, confirming the internship domain, duration, and structure.



Ref No: C16Y2026M051381655

Internship Offer Letter

Student Name : Mohd Kaif
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Python Full Stack Development + Project
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU67e4d13c8a5c71743049020

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Introduction to Full Stack & Python Fundamentals	Introduction to full-stack development concepts and environment setup. Learn Python fundamentals including data types, functions, OOP concepts, and backend basics.
Week 2	FastAPI & Data Validation (Pydantic)	Build backend APIs using FastAPI framework. Learn request handling, routing, dependency injection, and data validation using Pydantic.
Week 3	React Fundamentals & Hooks	Learn React basics including components, JSX, props, state management, useState, useEffect, and building dynamic user interfaces.
Week 4	Reusable Components & Frontend-Backend Integration	Create reusable React components and integrate frontend with FastAPI backend using API calls and dynamic data rendering.
Week 5	Database Integration with SQLAlchemy	Integrate databases using SQLAlchemy ORM, perform CRUD operations, manage models, and connect FastAPI with relational databases.
Week 6	Advanced React & State Management	Learn advanced React patterns, routing with React Router, and state management using Redux or Zustand for scalable applications.
Week 7	Testing & Deployment	Implement backend and frontend testing strategies. Learn deployment techniques using Docker and cloud platforms.
Week 8	API design & Front end optimisation Capstone Project – Full Stack Application Development & Capstone Project	Begin building a complete full-stack application (E-commerce platform) integrating FastAPI backend, database (SQLAlchemy), authentication system, and React frontend.

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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Table 4.3 records a brief weekly log of work completed during the internship.

Table 4.3: Weekly Progress / Internship Log

Week	Work Completed	Outcome
1	Set up development environment; revised Python fundamentals and OOP concepts.	Environment ready; Python basics reinforced.
2	Built first FastAPI routes; learned Pydantic-based request/response validation.	Working REST endpoints with validation.
3	Learned React components, JSX and hooks; built static UI screens.	Basic React UI components created.
4	Connected React frontend to FastAPI endpoints; built reusable components.	End-to-end data flow between frontend and backend.

Week	Work Completed	Outcome
5	Integrated SQLAlchemy models; implemented CRUD operations on the database.	Persistent storage for users and products.
6	Added Redux/Zustand state management and React Router navigation.	Multi-page app with consistent global state.
7	Wrote unit tests; explored Docker-based deployment.	Verified functionality; containerised build.
8	Built the capstone e-commerce module and the In-Memory Cache with TTL Lite project; final assessment.	Completed capstone project and mini-project; internship concluded.

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